



LESSON 3.1

POWERS OF 10

LESSON INTRODUCTION

MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

LEARNING TARGETS

- I can use laws of exponents to evaluate numerical expressions with a base of 10 and exponents greater than or equal to 0.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole number exponents and rational number bases.

WORKING TOWARD

MA.912.NSO.1.1 Extend previous understanding of the laws of exponents to include rational number exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

ESSENTIAL QUESTIONS

- How do you use the laws of exponents to evaluate numerical expressions with a base of 10?
- Which law of exponents did you use?

LESSON NARRATIVE

Students extend their prior knowledge of the laws of exponents to evaluate numerical expressions with bases of 10 and natural number exponents. Students begin by expressing the value of powers of 10 and identify the place value of the one. Students first guess the value of expressions, then check their work with calculators, and finally discover how to apply the laws of exponents.

COMPONENT SUMMARY

3.1.1 WARM-UP (~5 MIN.)

Explore applications of percentages and write large numbers

3.1.3 TRY IT (5 MIN.)

Guess values of powers of ten expressions

3.1.5 REFRESHER (10 MIN.)

Review laws of exponents

3.1.7 WRAP-UP (~5 MIN.)

Reflect on mastery of learning target

3.1.2 EXPLORATION (10 MIN.)

Explore powers of ten and place value

3.1.4 EXPLORATION (10 MIN.)

Use calculator to check values of powers of 10 expressions

3.1.6 GUIDED PRACTICE (5 MIN.)

Identify the laws of exponents used to evaluate numerical expressions

RESOURCES

GLOSSARY

Key Terms:

- Product of Powers Exponent Law
- Power of a Product Exponent Law
- Quotient of Powers Exponent Law
- Identity Exponent Law
- Zero Exponent Law

Additional Terms:

- N/A

MATERIALS

- Calculator
- (Optional) Highlighter

ADDITIONAL PREPARATION

- Consider adding the laws of exponents to the classroom Word Wall or creating Anchor Charts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may think there is an exponent law of adding and subtracting powers of 10.
- Students may count the one, instead of only the zeros, when they express the numbers using a power of 10.

THINGS TO CONSIDER

- Students may be overwhelmed by the large numbers. Discuss that this is why we use exponents to express the numbers.
- If time is an issue, then consider assigning 3.1.6 Guided Practice for homework.
- Determine if your students need the support of all or part of the 3.1.5 Refresher. The students reviewed laws of exponents in unit 1. The Refresher is another opportunity to review the laws.

LITERACY AND LANGUAGE ROUTINES

Poll the Class

Consider polling the class about which law of exponents they identified in each expression in 3.1.6 Guided Practice. Ask students to justify their answers and critique the reasoning of others.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

In 3.1.4 Exploration and 3.1.5 Refresher prompt students to communicate their ideas and methods effectively. Suggest that students ask questions of their peers and compare methods and justifications.

DIFFERENTIATE INSTRUCTION

In 3.1.5 Refresher, provide a visual entry point by having students highlight, circle, or underline the parts of the expressions that were impacted by the laws of exponents. Display the laws of exponents for students to reference. Provide an auditory entry point by asking students to read the laws of exponents in 3.1.5 Refresher aloud.

Which is an example of the **quotient of powers** exponent law?

- ☐ $\frac{a^m}{a^n} = a^{m-n}$
- ☐ $\frac{a^m}{a^n} = (a - a)^{m-n}$
- ☐ $\frac{a^m}{a^n} = 1^{m-n}$

3.1.1 WARM-UP: GRAINS OF SAND

TEACHER MOVES

Florida has a wide range of sands. Consider using this opportunity to have students share their life experiences with the different types of sands – from beaches to their backyards. This may give students an opportunity to share something of interest to them.



3.1.1 Warm-Up: Grains of Sand

The beaches on Siesta Key are 99% quartz, meaning that the beaches are made of some of the finest grains of sand.



- Suppose you have a teaspoon of the sand from Siesta Key. How could the teaspoon of sand be used to estimate the number of grains of sand in a bucket of sand?

Answers will vary. If you count the number of grains of sand in a teaspoon, and then figure out how many teaspoons are in the bucket, you can multiply the number of grains in a teaspoon by this number to determine the number of grains in the whole bucket.

A group of researchers at the University of Hawaii used a teaspoon of sand to estimate the number of grains of sand on all of the beaches on the Earth. Their estimate was that there are roughly seven quintillion, five hundred quadrillion grains of sand. A quintillion is 1,000,000,000,000,000 and a quadrillion is 1,000,000,000,000.

- Write seven quintillion, five hundred quadrillion.
7,500,000,000,000,000,000



3.1.2 Exploration: Powers of Ten

1. Use a scientific calculator to find the value of each power of 10.

Power of 10	Value
10^1	10
10^2	100
10^3	1,000
10^4	10,000
10^5	100,000
10^6	1,000,000
10^7	10,000,000
10^8	100,000,000
10^9	1,000,000,000
10^{10}	10,000,000,000

2. What do you notice about the relationship between the power of 10 and its value?

Answers will vary. The 1 is the same number of places from the left as the value of the exponent.

3. Work with your partner to complete the table.

Power of 10	Value	Place Value of 1
10^1	10	Tens
10^2	100	Hundreds
10^3	1,000	Thousands
10^4	10,000	Ten Thousands
10^5	100,000	Hundred Thousands
10^6	1,000,000	Millions

4. Work with your partner to describe how the exponent of the power of ten relates to place value.

Answers will vary. There are the same number of zeroes as the exponent.

5. Based on the pattern in the table, what is the value of 10^0 ?

1

3.1.2 EXPLORATION: POWERS OF TEN

TEACHER MOVES

Exploring the laws of exponents with powers of 10 will allow students to work with relationships of place value to solidify their understanding of the laws of exponents.

TEACHER MOVES

Review place values and have students create a reference sheet or foldable in their interactive notebooks. Ask students to share how they remember the place values.



3.1.3 Try It: Make a Guess

1. Without your partner or a calculator, make a guess on the value of each mathematical statement. Do not worry about being wrong. Remember, mistakes are a part of learning.

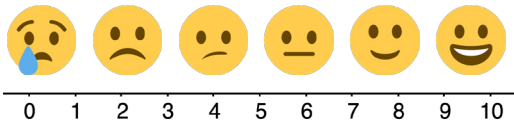
Whenever possible, write your guess using a power of 10. Show your work or explain your thinking.

Mathematical Statement	My Guess	My Work or Explanation
$10^6 + 10^7$	11,000,000	$10,000,000 + 1,000,000$
$10^2 \times 10^6$	10^8	10^{2+6}
$10^5 - 10^3$	99,000	$100,000 - 1,000$
$\frac{10^6}{10^4}$	10^2	10^{6-4}
$(10^4)^2$	10^8	$10^{4 \times 2}$
$\frac{1}{10^2}$	10^{-2}	10^{0-2}

2. How confident are you in your guesses?

Use the emoji scale to rate your confidence in your guesses.

Answers will vary.



3.1.3 TRY IT: MAKE A GUESS

TEACHER MOVES

This is an opportunity to allow students to explore without judgement how the values will change based on the operation and the exponents. Create a safe environment for students to freely make guesses and mistakes. Remind them that a growth mindset embraces challenges and mistakes.

3.1.4 EXPLORATION: USING A CALCULATOR TO CHECK

QUESTIONING

Consider modeling how to enter exponents in calculators for students. Having all students use the same make and model calculator will eliminate potential confusion about how to enter the expressions. For example, you could use the Desmos scientific calculator.

TEACHER MOVES

Prompt students to discuss which mathematical statements can be rewritten using a power of 10 and which cannot. This discussion will help students realize when they can and cannot apply the laws of exponents to simplify expressions.

DIFFERENTIATE INSTRUCTION

Consider having students go back to the emoji scale to rank their new understanding based on using the calculator and talking with their partner. Consider providing one-on-one support for students who rated their confidence as a five or less the second time.



3.1.4 Exploration: Using a Calculator to Check

1. Use a calculator to complete the following.

- a. Find the value of each mathematical statement. Then, if possible, rewrite the value as a power of 10.

Mathematical Statement	Value Found Using the Calculator	Value Written as Power of 10
$10^6 + 10^7$	11,000,000	not possible
$10^2 \times 10^6$	100,000,000	10^8
$10^5 - 10^3$	99,000	not possible
$\frac{10^6}{10^4}$	100	10^2
$(10^4)^2$	100,000,000	10^8
$\frac{1}{10^2}$	0.01	10^{-2}

- b. Compare your guesses from the Try It to the values you found for each expression using a calculator.
- c. Discuss with your partner which mathematical statements could be rewritten using a power of 10 and which could not be. Summarize your findings.

Adding and subtracting with different exponents could not be rewritten. Multiplication and division could be rewritten using a power of ten.

**3.1.5 Refresher: Laws of Exponents**

The Laws of Exponents may help explain why some of the mathematical statements could be rewritten using a power of 10.

**Product of Powers Exponent Law**

When multiplying two powers that have the same nonzero base, keep the base and add the exponents.

- Which is an example of the product of powers exponent law?
 - $(a^m)^n = a^{mn}$
 - $(a^m)^n = a^{m+n}$
 - $a^m \cdot a^n = a^{mn}$
 - $a^m \cdot a^n = a^{m+n}$

**Power of a Power Exponent Law**

When raising a power to a power, keep the base and multiply the exponents.

- Which is an example of the power of a power exponent law?
 - $(a^m)^n = a^{mn}$
 - $(a^m)^n = a^{m+n}$
 - $a^m \cdot a^n = a^{mn}$
 - $a^m \cdot a^n = a^{m+n}$

**Quotient of Powers Exponent Law**

When dividing two powers that have the same nonzero base, keep the base and subtract the exponents.

- Which is an example of the quotient of powers exponent law?
 - $\frac{a^m}{a^n} = a^{m-n}$
 - $\frac{a^m}{a^n} = (a - a)^{m-n}$
 - $\frac{a^m}{a^n} = 1^{m-n}$

**Identity Exponent Law**

When raising a base that is a nonzero real number to the power of 1, the base is a factor one time, thus, has the same value as the base.

- Write an example of the identity exponent law.
 $a^1 = a$

**Zero Exponent Law**

When a nonzero real number is raised to the zero power, a^0 , its value equals 1.

- Write an example of the zero exponent law.
 $a^0 = 1$

3.1.5 REFRESHER: LAWS OF EXPONENTS**DIFFERENTIATE INSTRUCTION**

Provide a visual entry point by having students highlight, circle, or underline the parts of the expressions that were impacted by the laws of exponents. Display the laws of exponents for students to reference. Provide an auditory entry point by asking students to read the laws of exponents aloud.

TEACHER MOVES

For students who do not understand why the zero exponent law is true, use the Quotient of Powers Exponent Law to show them an example such as: $\frac{2^5}{2^5} = \frac{32}{32} = 1$ and that $\frac{2^5}{2^5} = 2^{5-5} = 2^0$ so $2^0 = 1$.

3.1.6 GUIDED PRACTICE: ALIGNING POWERS OF 10 STATEMENTS WITH THE LAWS OF EXPONENTS

TEACHER MOVES

Poll the Class

Consider polling the class which law of exponents they identified in each expression. Ask students to justify their answers and critique the reasoning of others.

DIFFERENTIATE INSTRUCTION

If students are struggling to count the zeros in question 2, prompt them to use the sets of commas, highlight, or cross out each zero as they count it. This method will provide a visual entry point.



3.1.6 Guided Practice: Aligning Powers of 10 Statements with the Laws of Exponents

- Determine which of the mathematical statements align with a law of exponents. Write the law in the table. If no law applies, then write "none."

Mathematical Statement	Law of Exponents
$10^6 + 10^7$	<i>none</i>
$10^2 \times 10^6$	<i>Product of Powers</i>
$10^5 - 10^3$	<i>none</i>
$\frac{10^6}{10^4}$	<i>Quotient of Powers</i>
$(10^4)^2$	<i>Power of a Power</i>
$\frac{1}{10^2}$	<i>Quotient of Powers</i>

- Numbers like a quintillion, 1,000,000,000,000,000,000, and a quadrillion, 1,000,000,000,000,000, are often used by scientists. When using such large numbers, scientists have developed a shortened method of writing these numbers called scientific notation. Scientific notation uses the powers of 10 to represent these numbers.
 - Write 1,000,000,000,000,000,000 using a power of 10.
 10^{18}
 - Write 1,000,000,000,000,000 using a power of 10.
 10^{15}



3.1.7 Wrap-Up: Reflection

1. Choose the question and sign that best describes your understanding of today’s lesson.

Answers will vary.

Are you confused?	Are you in the school zone?	Are you ready for the highway?
		

2. Briefly explain why you feel this way.

Answers will vary but should reflect students’ feelings about their learning in the lesson.

3.1.7 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Consider providing sentence stems or answer options for question 2.